```
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#Section: A
import numpy as np
import matplotlib.pyplot as plt
```

```
# Dataset
x = np.array([100, 150, 200, 250, 300])
y = np.array([20, 25, 30, 35, 40])

print("Input X:", x)
print("Input Y:", y)
```

```
Input X: [100 150 200 250 300]
Input Y: [20 25 30 35 40]
```

```
# Mean
mean_x = np.mean(x)
mean_y = np.mean(y)

print("Mean of X:", mean_x)
print("Mean of Y:", mean_y)
```

```
Mean of X: 200.0
Mean of Y: 30.0
```

```
# Calculate slope
numerator = np.sum((x - mean_x) * (y - mean_y))
denominator = np.sum((x - mean_x) ** 2)

b1 = numerator / denominator
print("Slope (b1):", b1)
```

```
Slope (b1): 0.1
```

```
# Calculate intercept
b0 = mean_y - b1 * mean_x
print("Intercept (b0):", b0)

# Prediction
y_pred = b0 + b1 * x

print("Original Y:", y)
print("Predicted Y:", y_pred)

# Residuals
residuals = y - y_pred

print("Residuals:", residuals)

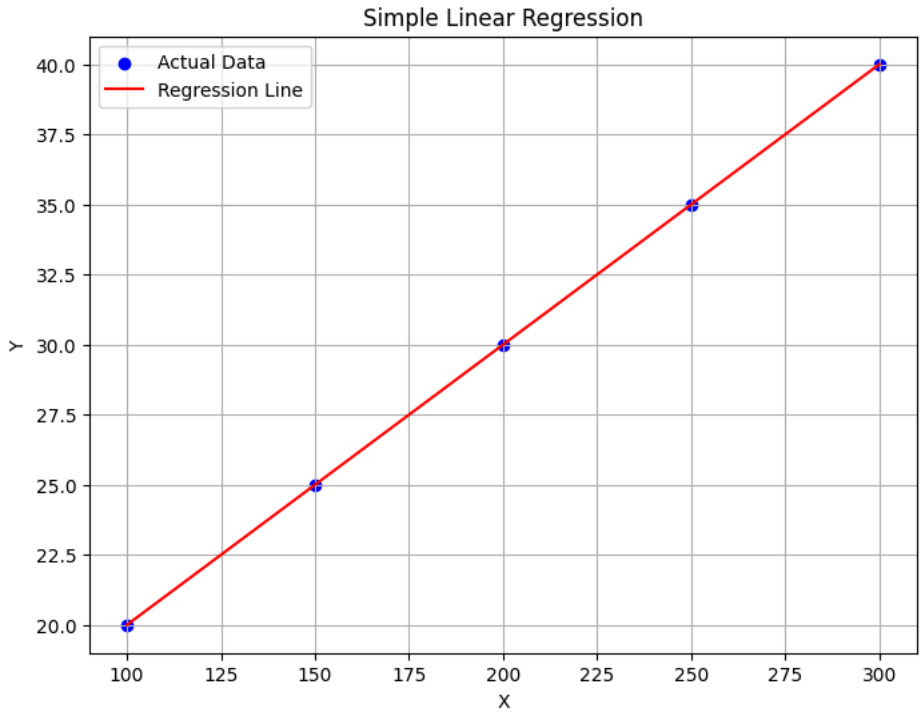
# Error Metrics
mae = np.mean(np.abs(residuals))
mse = np.mean(residuals ** 2)
rmse = np.sqrt(mse)

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
```

```
Intercept (b0): 10.0
Original Y: [20 25 30 35 40]
Predicted Y: [20. 25. 30. 35. 40.]
Residuals: [0. 0. 0. 0. 0.]
MAE: 0.0
MSE: 0.0
RMSE: 0.0
```

```
# Plot
plt.figure(figsize=(8,6))
```

```
plt.scatter(x, y, color='blue', label='Actual Data')
plt.plot(x, y_pred, color='red', label='Regression Line')
plt.title("Simple Linear Regression")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.grid(True)
plt.show()
```




```
Start coding or generate with AI.
```